ADDITIVE MANUFACTURING FOR IRRIGATION MEASUREMENT: FIELD-SCALE 3D PRINTED FLUMES FOR ACCURATE OPEN-CHANNEL FLOW MONITORING

HIGHLIGHTS

- Large 60° V-notch trapezoidal flume fabricated using large-format Fused Deposition Modeling (FDM) additive manufacturing
- Hydraulic performance validated against an industry standard fiberglass control and theoretical rating curve
- Hierarchical Bayesian analysis indicated no statistical difference between the 3D printed and stand flume discharge
- Field trails verify the structural durability and ease of installation for agricultural water monitoring applications

ABSTRACT. *Accurate open channel flow measurements are critical for agricultural water management, yet traditional fiberglass and metal flumes present significant financial and logistical challenges for scalable implementation. This study evaluated the hydraulic, structural, and economic viability of a full-scale, parametrically designed, large 60° V-notch trapezoidal flume fabricated with a large volume, Fused Deposition Modeling (FDM) 3D printer. A laboratory comparison flow experiment coupled with a hierarchical Bayesian stage-discharge model indicated no statistical difference in the hydraulic discharge characteristics between printed flumes and industry standard fiberglass. Field deployment of the printed flume in a furrow irrigated field, capturing five irrigation events, demonstrated the structural durability and dimensional stability (±2 mm tolerance) of the print material over two months of typical field conditions. Additive manufacturing reduced costs significantly over commercial flume units, allowing for rapid fabrication and scaling in applied settings. These findings demonstrate large-scale FDM as a highly accurate, adaptable, and cost-effective method for scaling small to medium-scale water monitoring infrastructure.*

Keywords. *Additive manufacturing, large format 3D printing, Hydraulic calibration, Irrigation monitoring, Hierarchical Bayesian analysis, Flume*

Accurate measurement of open-channel water flow is a foundational requirement for irrigation

DRAFT

management, water accounting, and water-quality monitoring across agricultural and rural landscapes. In the Western United States, where water rights determine who may divert, store, or use water at any given time and place, precise flow measurement is especially critical for ensuring legal compliance, equitable distribution, and defensible record-keeping. In the early twentieth century, Ralph Parshall, working jointly with the Colorado State University (CSU) Agricultural Experiment Station and the United States Department of Agriculture (USDA) Irrigation Investigations Unit, developed and calibrated what became one of the most widely recognized flow-measurement devices in the world, the eponymous Parshall flume (Parshall, 1921). For more than a century, flumes have functioned as reliable engineered structures for open-channel flow measurement in canals, ditches, and laterals. Their widespread adoption reflects not only their accessibility, but also their ability to provide consistent, predictable flow measurements across a wide range of operating conditions.

Despite their proven performance, flumes produced using traditional material such as fiberglass, steel, concrete, or aluminum, present economic and practical barriers to widespread deployment for field-scale and small watershed monitoring (Harmel et al., 2023). Commercially produced flumes sized for flows of approximately 3 cubic feet per second (CFS), a common capacity for irrigation laterals in the Western U.S., typically cost ($1,800 to more than $3,000 per unit), depending on materials and manufacturer. Additionally, their bulk size and weight incur significant shipping costs and introduce lead time constraints, especially for producers, conservation districts, and irrigation companies operating in remote areas. Alternatively, flumes are often constructed locally using concrete, which can present limitations in areas where site access, materials, and/or expertise are unavailable. Additionally, concrete flumes, although relatively affordable compared to fiberglass and metal flumes, are unable to be moved from one location to another, creating an additional barrier to scalable deployment. These financial and logistical constraints can limit broader adoption of flume infrastructure needed to support accurate accounting, equitable distribution, and more efficient water management. Locally manufactured alternatives that preserve hydraulic accuracy while reducing production costs could

DRAFT

dramatically expand access to measurement instruments essential for water stewardship across the U.S.

Additive manufacturing, more commonly known as 3D printing, offers a compelling pathway toward achieving this goal. Laboratory-scale studies demonstrate that printed flumes can be produced at very low cost while reproducing the internal geometries and accurate discharge relationships required for open-channel flow measurement (Akhter et al., 2021; Oertel and Shen, 2022). Akhter et al., (2021), for example, evaluated 0.122-m HS and 0.102-m Palmer–Bowlus flumes printed in polylactic acid (PLA) and polyethylene terephthalate glycol (PETG), showing remarkably consistent performance across materials and replications, with adjusted R^2 values exceeding 97% for all tested configurations. These results established the feasibility of 3D printing to reduce costs and enable rapid prototyping of small flumes for hydrologic monitoring. However, previous work has been limited to laboratory-scale devices printed on desktop-class printers (e.g.,256 mm by 256 mm by 256 mm), leaving an important gap in understanding whether additive manufacturing can be extended to monolithically-printed field-scale, field-ready flumes that match the performance, durability, and dimensional accuracy required for agricultural water management.

Current large format fused deposition modeling (FDM) printers now enable production of flumes exceeding 0.8 m in width and 1.0 m in height, raising the possibility of fabricating flumes at the scales used in irrigation ditches and laterals. Local 3D printing fabrication has the potential to reduce production costs significantly, eliminate shipping barriers, and allowing for rapid design customization through parametric (i.e., a design technique that involves creating 3D models using parameters, relationships, and constraints.) computer aided design (CAD) modeling.

Therefore, the goal of this study is to create field-scale 3D printed flumes using a large FDM printer and verify flume accuracy against commercial equivalents and to demonstrate the deployment of a 3D-printed flume *in-situ* in conjunction with an edge-of-field water quality monitoring apparatus. This work advances both the scientific and practical foundations of additive manufacturing by applying it to agricultural water management, thereby expanding access to accurate, accessible tools that strengthen

DRAFT

water stewardship.

Materials and Methods

Flume Design and Digital Modeling

A 3D CAD model of a large 60° V-trapezoidal flume was created using Autodesk Fusion 360 (Autodesk, 2024). This flume model was selected because it is heavily utilized in the Agricultural Water Quality Program's current research, providing accurate flow measurement capability in furrow irrigated systems around the state of Colorado. The design was based on dimension specifications from the Open Channel Flow Measurement Handbook (Teledyne ISCO, 2023) with the original design principles established by Robinson et al., (1960). A fully parametric modeling approach was utilized, where the physical geometry was constrained by defined mathematical variables and relational constraints rather than static dimensions. This allows for easy dimensional adjustments while maintaining critical geometric relationships as defined in the master V-trapezoidal flume schematic (Figure 1). Unlike fiberglass fabrications, which use static, expensive molds that lock designs in a set dimension, the parametric workflow allowed for the rapid modification of key geometric variables, such as the throat width (b_2), wall thickness, and sidewall slope (θ), to meet site specific needs without additional re-tooling cost. A primary advantage of digitally modeled and 3D printed trapezoidal flumes is the ability to prescribe the contraction and throat geometry (key metrics defined in Figure 1 as $L, b_2, W, F\ D,$ and θ) to intentionally force critical flow at the control section.

DRAFT

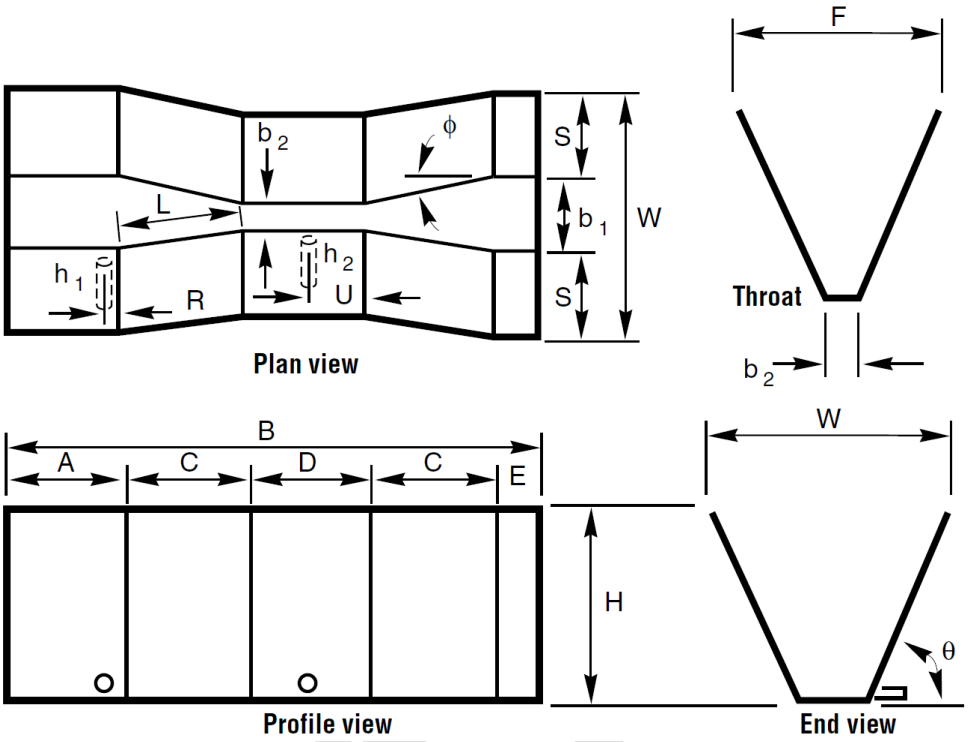


Figure 1. Master drawing of the large 60° V-notch trapezoidal flume from Open Channel Flow Handbook (Teledyne ISCO, 2023) as originally proposed by Robinson et al., (1960) used to create the CAD design for the 3D printed model.

The flume was designed as five separate components composed of a monolithic main body, a vertical support panel (acting as a leg), and three horizontal cross struts (Figure 2). The support panel and struts were printed separately to eliminate the use of 3D print supports (i.e., auto-generated structures to support the 3D printed models when overhangs exist in the printing orientation), reducing material uses and printing time. The flume body was modeled with a uniform 12 mm wall thickness and edges not in contact with the flow path were filleted (two mm) to reduce stress concentrations and print failure. A stilling well port was integrated into the design to accommodate a standard PVC pipe for water level instrumentation. Additionally, the screw holes to mount the horizontal cross struts and vertical supports panels were designed to align the parts while creating a stable structure.

DRAFT

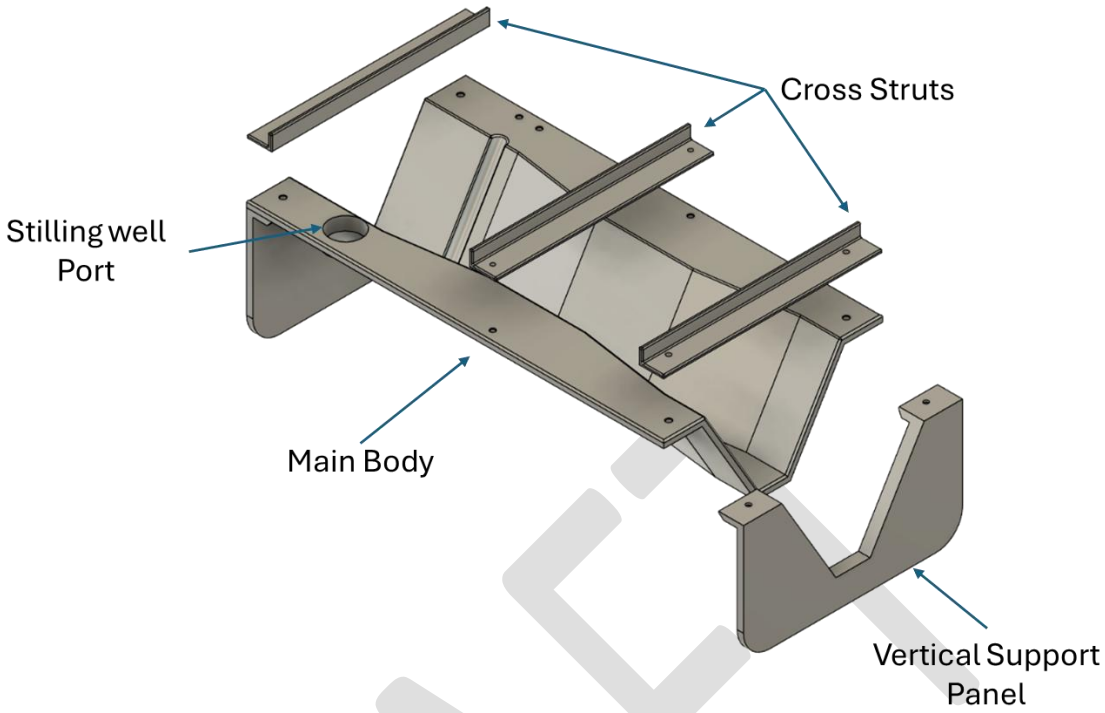


Figure 2. Exploded view of the 3D printed flume assembly from Autodesk Fusion. Component includes a monolithic main body with an integrated stilling well port, cross struts, and vertical support panel.

As with conventional critical-depth flumes, discharge is governed by the minimum specific-energy condition, obtained from $dE/dy = 0$, where E is specific energy of the flow, and y is the depth of flow, which gives

$$Q^2 = g\frac{A^3}{T} \quad (1)$$

where Q is discharge, g is gravitational acceleration, A is the cross-sectional flow area at the throat, and T is the corresponding top width (Chow V.T., 1959). For the trapezoidal throat geometry, these quantities are determined directly by the bottom width b_2, sidewall slope (implied in θ), and flow depth, Under free-flow conditions, the throat section controls discharge such that stage measured at a fixed upstream location R defines Q through geometry and gravity alone, with non-ideal effects represented by discharge coefficients. While this study evaluates a standard configuration, the ability to modify these geometric components within a digital design environment enables theoretically guided

DRAFT

customization across a range of canal sizes and flow capacities without requiring new commercial fabrication.

Printing Process

Gcode print files were prepared using OrcaSlicer v2.3.0 (SoftFever, 2025). Fabrications were performed using an Elegoo OrangeStorm Giga FDM printer (Elegoo, Shenzhen, China (~$2,500)). This system features an 800 x 800 x 1000 mm (LxWxH) build volume, which allowed the long-throated sections of the flumes to be printed as a single part, minimizing joints that could disrupt hydraulic flow.

Two flumes were printed to have the ability to compare printing consistency and subsequent hydrological property impacts. The flumes were printed using a standard 0.6 mm brass nozzle with parameters optimized for structural integrity and watertight performance. PETG filament (SUNLU, Zhuhai, China) was selected as the printing material. PETG was chosen for its dimensional stability, waterproof, resistance to ultraviolet (UV) degradation, and high heat resistance without the need of a printer enclosure. A brim was applied to the base of the print to ensure bed adhesion and mitigate warping. These prints were conducted without an enclosure. Complete printer settings are shown in Table 1.

To print each flume, the main body was oriented vertically to utilize the full Z-axis height (1000 mm) of the printer (Figure 3). After the main body was completed, it was removed from the printer build plate, then cross struts and vertical support panel were printed separately. The total material used for this complete flume assembly was approximately 6.34 kg of filament (material cost ~$90) and took approximately 78 hours to print.

DRAFT

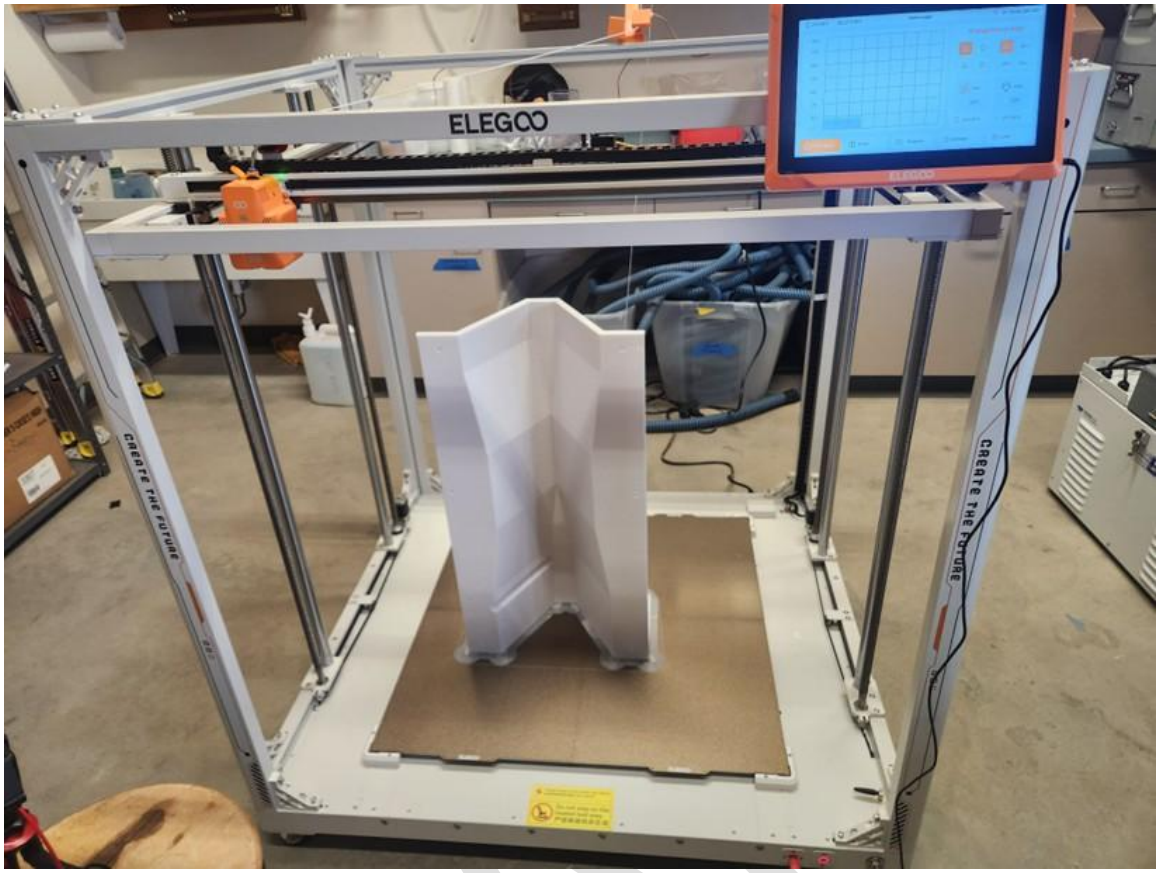


Figure 3. An Elegoo OrangeStorm Giga, large-format industrial FDM 3D printer with a large 60° V-trapezoidal 3D printed flumes on the bed

Table 1. Printing parameters for the large 60 ° V-trapezoidal 3D printed flume

Parameter	Setting
Material	PETG (SUNLU)
3D Printer	Elegoo OrangeStorm Giga
3D Printer Accuracy	0.1mm
Slicer Software	OrcaSlicer
Nozzle Diameter	0.6 mm
Layer Height	0.36 mm
Wall Thickness	5 perimeters (i.e., 30 mm)
Infill Density	25% (Cubic pattern)
Nozzle Temperature	240 C
Bed Temperature	70 C
Printing Speeds	Outer wall: 150 mm/s; Infill 250 mm/s
Acceleration	2500 mm/s^2
Bed Adhesion	15 mm Brim
Cooling Fan	0% (layer 1), 10-40% (rest of layer)

DRAFT

Post-Processing and Assembly

After each print was complete, a deburring tool was used to manually remove the plastic brims from the bottom edge of the flume components. Exterior seams were wet sanded with P320 grit sandpaper, ensuring that the interior surface was left untouched to evaluate the hydraulic performance of the standard printing surface roughness with defined print settings.

Assembly was performed using a combination of chemical welding and mechanical fastening. The Front vertical support panel, along with one cross strut, was secured to the main flume body using 3D Gloop (St. Louis, MO, USA) adhesive and two stainless steel ¼ -inch coarse-thread bolts and nuts. Next, the remaining two cross struts were bolted into position, one in the middle and rear end.

To accommodate water level instrumentation during field deployment, a still well was assembled using a standard 2-inch diameter, 18-inch long PCV pipe. This pipe was bounded directly to the stilling well port located on the top of the main flume body and an end cap on the bottom end using 3D Gloop to ensure watertight seal. A connection was made from the bottom of the flume to the still well was made using flex tube, a pipe clamp, and 3D Gloop to secure it. The stilling well was positioned to extend 9-inches below the flume bottom (zero mark), allowing suspended solids to settle out of the measurement column ensuring accuracy when measuring water level.

Hydraulic Calibration

Hydraulic calibration/testing of each constructed flume was conducted at the Colorado State University Hydraulics Laboratory (CSU; Fort Collins, CO) under controlled conditions. Both 3D printed flumes, and one commercial fiberglass flume of the same model, were tested individually by a recirculating open-channel test section (Figure 4a). Each flume was installed and enclosed such that all discharge measured in the test section passed through the trapezoidal flumes (Figure 4b and Figure 4c).

DRAFT

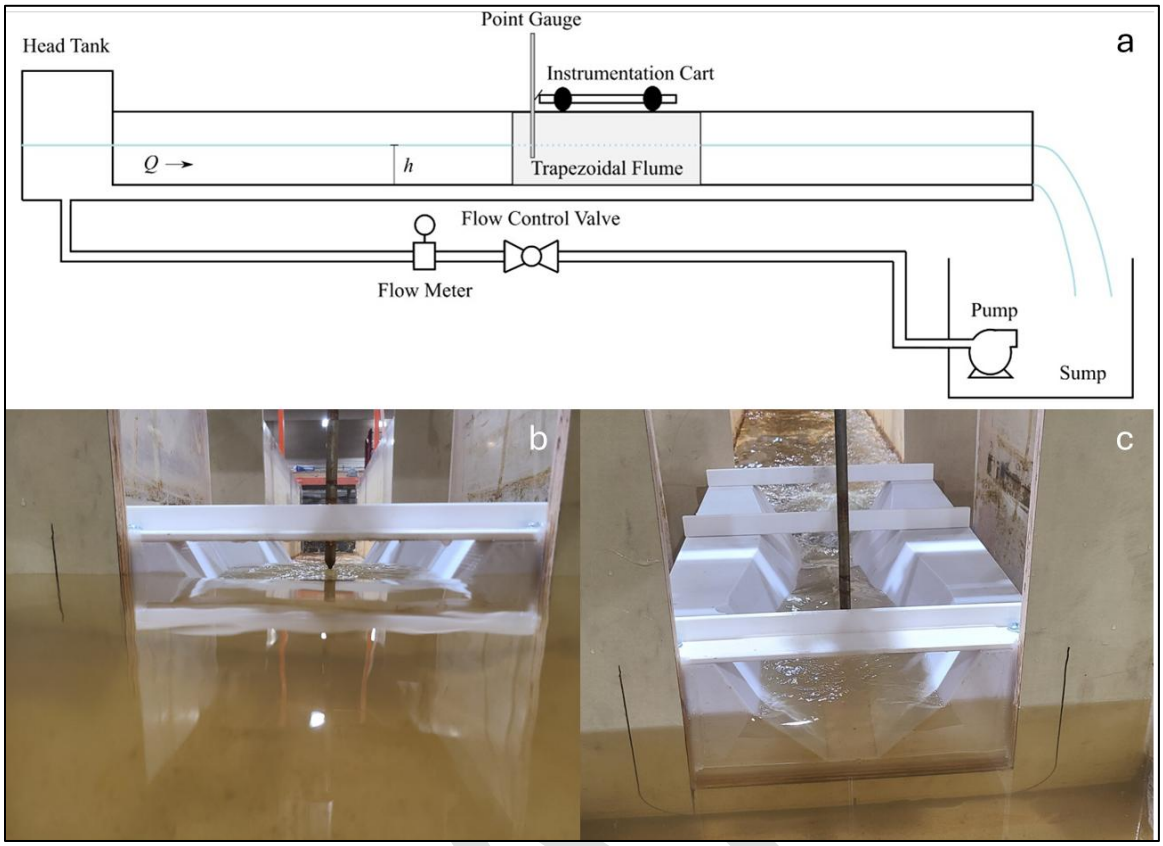


Figure 4. (a) Schematic depicting hydraulic testing conditions of each trapezoidal flume evaluated in test channel. (b) Front view and (c) isometric view of a 3D printed Large 60-degree-V trapezoidal flume during free-flow testing.

The test section consisted of a rectangular open channel measuring 0.61 m wide, 0.61 m deep, and 18.3 m long. Discharge was supplied by a 75 hp variable-speed pump and regulated through the laboratory flow-control system of globe valves. Volumetric flow rate was measured using an inline Signet 2550 insertion magnetic flow meter (Signet Scientific Company, Monte, CA, USA.) with a stated accuracy of $\pm 2.0\%$ of the reading. Water-surface elevations were measured using a vertically adjustable stainless-steel rod mounted to a fixed reference frame. Elevations were read relative to a metric steel rule with 1 mm graduations. Stage (i.e., depth) measurement uncertainty (± 1.5 mm) reflects the combined effects of reading precision, capillary deformation of the free surface at the rod-water interface, and minor dynamic surface distortion during flow (Figure 4b and Figure 4c). Flume infiltration was assumed to be ± 0.02 CFS, based on measurements obtained upstream and downstream of the test section during CSU hydraulic facility commissioning. Device measurement

accuracies are summarized in Table 2.

Table 2. Equipment accuracies reported by their respective manufacturer manuals.

Equipment	Accuracy
Magnetic flow meter	+/- 2% of reading
Point gauge	+/- 1.5 (mm)
Flume infiltration	+/- 0.02 cfs

For each flume, a reference point-gauge reading at the flume bottom was used to establish a consistent datum. Flow depth, h, was computed as the difference between the measured water surface elevation and the bottom reference elevation. Stage-discharge measurements were collected across the operational flow range of each flume under steady-flow conditions. Stage-discharge pairs were measured across the operational range of each flume. Stage-discharge relationships were then parameterized using a power-law equation of the form (Eq. 2) as established in Robinson and Chamberlain (1960):

$$Q = ah^b \tag{2}$$

where Q is discharge, h is flow depth, and a and b are empirical coefficients. Accordingly, the manufacturer's default parameter values are 1.55 and 2.58 for a and b, respectively (Teledyne ISCO, 2023). These coefficients are reported to provide ± 2–5% accuracy under free-flow conditions up to approximately 80% submergence. All stage-discharge measurements used for calibration were collected under free-flow (unsubmerged) conditions; downstream tailwater levels were maintained sufficiently low to prevent submergence effects, ensuring that the rating relationships reflect true free-flow behavior consistent with other V-notch trapezoidal flume manufacturer specifications. These fitted relationships were used for subsequent statistical analysis and comparison among flume types.

Bayesian Stage–Discharge Comparison and Uncertainty Quantification

Eq. 2 shows the model form for the expected stage-discharge relationship in the trapezoidal flumes used in this study. Model parameters a and b need calibration by using observations collected by the

DRAFT

magnetic flow meter and point gauge, while accounting for their respective device uncertainties and additional uncertainty incurred by potential flume infiltration (i.e., water infiltrating into flume media).

Rather than assuming the observed discharge (Q_{obs}) equals the true discharge (Q_{true}), a Bayesian approach was used to fit parameters to the unobserved Q_{true} (i.e., a latent parameter) by explicitly modeling the measurement system and hydraulic process separately. This formulation allowed uncertainty from multiple sources to propagate through to the estimated calibration curves and predicted discharges.

The hierarchical model (i.e., a statistical framework that analyzes multi-level data by organizing parameters into nested layers such as flume type and latent flow) ultimately predicts Q_{obs} (i.e., the estimand) as a function of Q_{true} (i.e., the estimate), with prior distributions assumed that were updated by observed data. Observed discharge was modeled as a noisy realization of the latent true discharge (Eq. 3):

$$Q_{\text{obs},i} \sim \text{Normal}\left(Q_{true,i}, \sqrt{(c_v Q_{\text{true},i})^2 + \sigma^2_{Q,\text{add},j}}\right) \quad (3)$$

where $Q_{\text{obs},i}$ is the observed discharge associated with observation i, and $Q_{\text{true},i}$ is the latent true discharge, both measured in CFS. Additionally, this heteroscedastic formulation includes two physically motivated error components:

1. Proportional measurement error, represented by a global (i.e., over all measurements) coefficient of variation, c_v, consistent with flow-meter precision expressed as a percentage of reading.
2. Additive discharge loss, $\sigma_{Q,\text{add},j}$, represented flume infiltration losses that are independent of flow magnitude for each flume tested, j.

Combining these terms allowed uncertainty to scale appropriately with discharge.

For each observation i, the latent true discharge was modeled as shown in Eq. 4:

$$Q_{\text{true},i} \sim \text{Normal}(\mu_i, \sigma_{\text{proc},j}) \quad (4)$$

With the estimator defined as stated in Eq. 1, updated with flume and observation notation (Eq. 5):

DRAFT

$$\mu_i = a_j \, H_i^{b_j} \tag{5}$$

The process variance $\sigma_{\text{proc},j}$represents small, irreducible fluctuations in flow at a given stage, including the point gauge uncertainty, coupled with other potential unobserved effects for each flume separately.

Calibration parameters were estimated separately for each flume, while sharing common prior structure based on the published manufacturer equation parameters for the tested flume design (Openchannelflow, Dahlonega, GA, USA):

$$a_j \sim \text{LogNormal}(\log(1.55), 0.5) \tag{6}$$

$$b_j \sim \text{LogNormal}(\log(2.58), 0.5) \tag{7}$$

$$\sigma_{\text{proc},j} \sim \text{Exponential}(1) \tag{8}$$

$$\sigma_{Q,\text{add},j} \sim \text{LogNormal}(\log(0.2), 0.5) \tag{9}$$

The proportional error parameter was shared across flumes:

$$c_v \sim \text{Lognormal}(\log(2 \times 10^{-4}), 0.5) \tag{10}$$

The priors on a_j and b_j were centered on the manufacturer's published trapezoidal flume equation, providing a realistic hydraulic baseline while allowing deviation. The lognormal form enforces positivity and reflects prior uncertainty spanning approximately an order of magnitude (i.e., 0.5). A relatively uninformative, exponential prior was used for each $\sigma_{\text{proc},j}$, also enforcing positivity in values. The prior for the proportional error $c_v$was centered on manufacturer-reported flow-meter precision, with sufficient spread to accommodate deviations due to installation or operating conditions. Additive infiltration loss priors were centered on small but non-zero values. Prior parameters were log-transformed because it improved computational stability and makes uncertainty propagation more robust across several orders of magnitude in flow. Importantly, these priors did not enforce agreement with the manufacturer's curve. Rather, they encoded physically reasonable expectations that prevent overfitting while allowing the posterior to depart meaningfully when warranted by the data.

Hamiltonian Monte Carlo (HMC) engine Stan v. 2.36 (Stan Development Team, 2023) was used to

DRAFT

estimate the posterior distribution for each parameter using CmdStanR (Gabry et al., 2025). Posterior distributions were processed in R v. 4.4.2. (R Core Team, 2020) using RStudio (Posit team, 2025), with the Statistical Rethinking R package, v. 2.42 (McElreath, 2020). Statistical models were validated with simulated data to ensure that they could recover the simulated parameters. Additionally, for all analyses, visual inspection of the trace plots, marginal HMC energy inspection, Gelman–Rubin diagnostic (Gelman and Rubin, 1992), and the effective number of samples (Vehtari et al., 2021) indicated model convergence (see supplementary Table S1 and Figure S2). Additionally, all R code used to reproduce these results can be found in Supplementary Material S3.

Field Installation and Applied Case Study

A field trial was conducted at the Colorado State University Agricultural Research, Development and Education Center (ARDEC) in Fort Collins, CO, to evaluate structural durability and performance of one 3D printed flume. The unit was installed in a furrow irrigated, 3-hectare corn field near the end of the field in an edge-of-field monitoring configuration (Figure 5). The flume was installed in an irrigated furrow with the flume floor flush with the upstream and downstream furrow ground to allow free flow of irrigation water through the flume. To prevent shifting during flow events, the flume was secured with soil and compacted using a hand tamper to lock the unit in place. To prevent flow bypass, soil berms were constructed on the upstream right and left side of the flume into adjacent furrow banks.

A CSU-developed low-cost water sampler (LCS; Deleon et al., 2025) was deployed alongside the 3D printed flume for continuous water quality sampling and monitoring during deployment. A Milone eTape fluid level sensor (Sewell, NJ, USA) was mounted within the attached PVC stilling well, allowing for water level measurement away from surface turbulence and debris. This level data, combined with the calibrated rating curve, allowed for calculation of total flow volume. Water samples were collected from the outflow section of the flume to ensure minimal disturbance to the hydraulic profile within the throat. The trial period was from August 4, 2025, to October 2, 2025, capturing five distinct irrigation events.

DRAFT

Following the conclusion of the field trial, the flume was uninstalled and inspected for damages and overall condition; critical dimensions, including throat width, and sidewall slope, were assessed to determine if hydraulic forces of environmental exposure had caused any deformations from the original geometry.

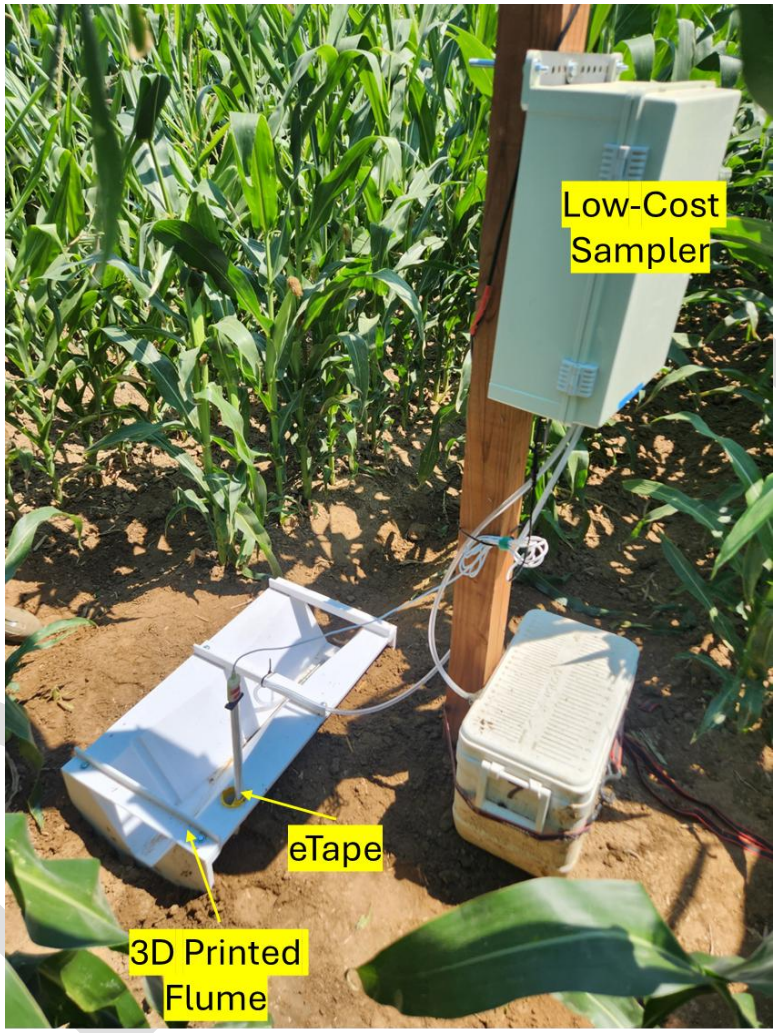


Figure 5. 3D printed 60° V-notch trapezoidal flume in a furrow irrigated field at the Agricultural Research Development and Education Center (ARDEC). Continuous stage monitoring by the eTape sensor installed in the PVC stilling well.

RESULTS AND DISCUSSION

STAGE–DISCHARGE RELATIONSHIPS AND PERFORMANCE COMPARISONS

Measured stage and discharge data used for calibration of the three trapezoidal flumes are shown in

Table 3. Reported flow depth was computed as the difference between the point gauge reading at the water surface and the corresponding bottom reference reading for each flume, yielding the effective hydraulic head used in subsequent analysis. The dataset includes measurements for two 3D printed trapezoidal flumes (named “3D Printed 1” and “3D Printed 2”) and one commercial fiberglass flume. Both 3D printed flumes had six stage-discharge measurements (n = 6) whereas the fiberglass flume had four measurements (n = 4).

Table 3. Raw laboratory calibration measurements for trapezoidal flumes, including point gauge readings, bottom reference elevations, computed flow depth (difference between gauge and bottom readings), and corresponding measured discharge. Measurements were collected for two 3D-printed flumes and one commercial fiberglass flume under steady-state flow conditions and were used as inputs to the Bayesian stage–discharge model.

Point Gauge Reading (ft)	Bottom Reading (ft)	Depth (ft)	Flow Rate (CFS)	Flume Type
2.737	3.108	0.371	0.10	3D Printed 1
2.695	3.108	0.413	0.15	3D Printed 1
2.651	3.108	0.457	0.20	3D Printed 1
2.612	3.108	0.496	0.25	3D Printed 1
2.58	3.108	0.528	0.30	3D Printed 1
2.565	3.108	0.543	0.32	3D Printed 1
2.861	3.102	0.241	0.03	3D Printed 2
2.774	3.102	0.328	0.08	3D Printed 2
2.711	3.102	0.391	0.13	3D Printed 2
2.668	3.102	0.434	0.18	3D Printed 2
2.621	3.102	0.481	0.23	3D Printed 2
2.581	3.102	0.521	0.28	3D Printed 2
2.047	2.373	0.326	0.04	Fiberglass
2	2.373	0.373	0.08	Fiberglass
1.96	2.373	0.413	0.12	Fiberglass
1.842	2.373	0.531	0.26	Fiberglass

Stage–discharge data collected for the two 3D printed trapezoidal flumes and commercial fiberglass trapezoidal flume exhibited the expected nonlinear power-law relationship characteristic of long-throated flumes. Across all devices, discharge increased monotonically with stage and fitted rating

DRAFT

curves closely followed measured observations (Figure 6).

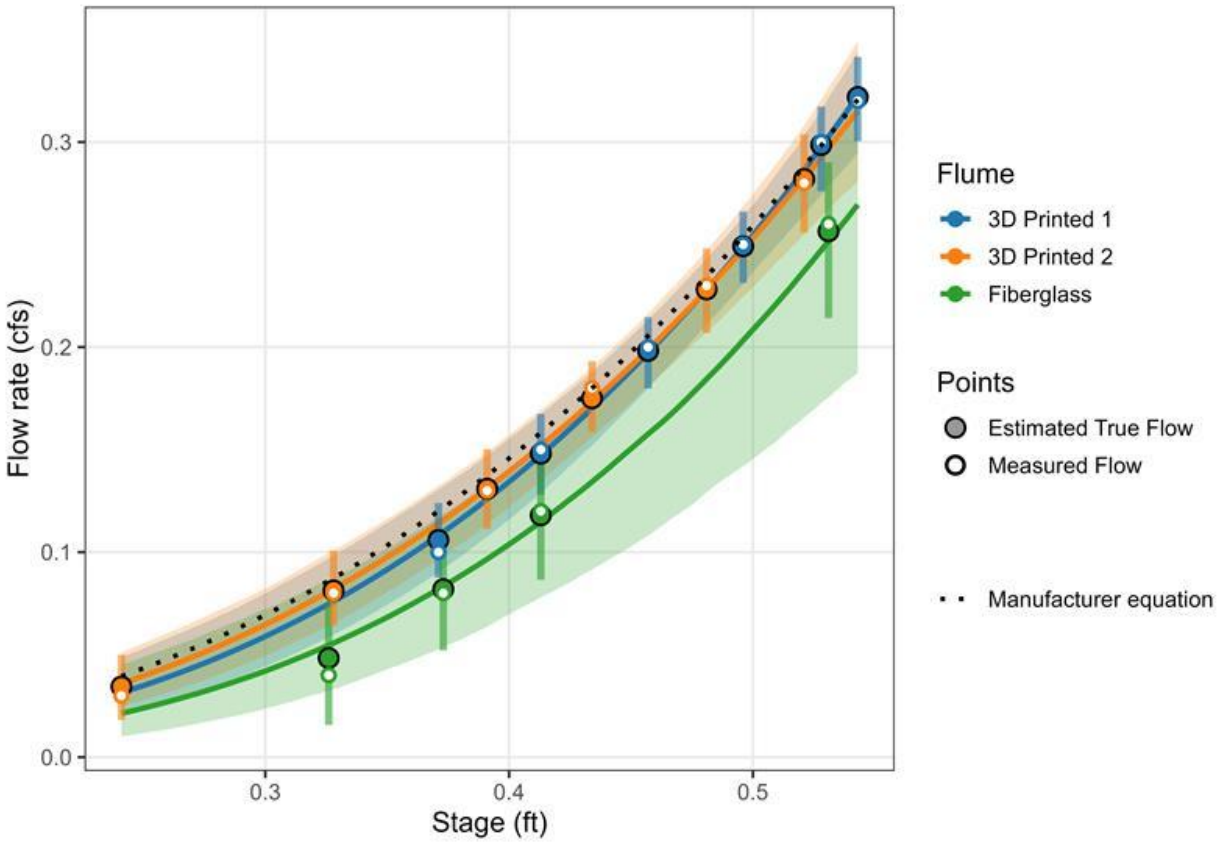


Figure 6. Bayesian stage–discharge calibration for two 3D-printed trapezoidal flumes and a commercial fiberglass trapezoidal flume. Solid colored lines show posterior median rating curves derived from the hierarchical Bayesian model, with shaded bands representing 95 % high density probability intervals for parameter uncertainty. Open circles denote measured discharge observations, while filled circles indicate posterior estimates of latent true discharge. The manufacturer's empirical stage–discharge equation is shown for reference (dotted line). Across the tested depth range, the 3D-printed flumes closely reproduce the hydraulic behavior of the fiberglass flume within overlapping credible intervals.

Posterior median rating curves derived from the hierarchical Bayesian model show that both 3D printed flumes track the manufacturer equation more closely than the fiberglass flume across most of the tested stage range (0.1–0.6 ft). At moderate to higher depths (0.45–0.55 ft), the 3D printed flumes nearly overlay the manufacturer curve, while the fiberglass unit consistently under-predicts discharge relative to the reference equation at all depths. This systematic deviation is reflected in both the curvature parameter (b) and the additive discharge offset terms (i.e., $\sigma_{Q,add}$) estimated by the model.

The posterior estimate of the proportional flow-meter error term (±0.022%) was lower than the manufacturer's specified magnetic flow meter precision (±2% of reading), suggesting that this experimental setup resulted in significantly less measurement error than what is expected over the full measurement range of the flow meter.

Uncertainty is summarized using Bayesian Highest Posterior Density Intervals (HPDI) rather than traditional confidence intervals. Parameter results and uncertainties are reported in Table 4. A 95% HPDI represents the range of parameter values that contains 95% of the posterior probability mass, given the observed data and model assumptions. In practical terms, this means that, conditional on the data collected during flume calibration and the specified model, there is a 95% probability that the true parameter value, such as a rating-curve coefficient in $Q = ah^b$, lies within this interval. This interpretation follows directly from the Bayesian framework, where uncertainty is expressed as probability over unknown quantities.

By contrast, a 95% confidence interval is defined in terms of repeated sampling. Specifically, if the same experiment, such as the stage–discharge testing described in this study, were repeated many times, 95% of the constructed intervals would contain the true parameter value. However, for any single dataset, the confidence interval does not permit a direct probabilistic statement about the parameter itself. The parameter is treated as fixed, and the interval is random, which makes interpretation less intuitive in applied settings.

DRAFT

Table 4. Posterior summaries of flume-specific and shared parameters from the Bayesian stage–discharge calibration model. Values are posterior means with 95% highest posterior density intervals (HPDIs). The parameters (a) and (b) describe the scale and curvature of the power-law rating curve, ($\sigma_{\mathbf{proc}}$) represents process variability, ($\sigma_{Q,\mathbf{add}}$) represents additive discharge loss, and (c_v) denotes proportional flow-meter error common to all flumes. Note that for the manufacturer's equation, a = 1.55 and b = 2.55.

Parameter	Description	3D Printed 1	3D Printed 2	Fiberglass
a	Scale term in power-law equation ($Q = a\,H^b$); shifts the curve vertically.	1.88 (1.35 – 2.42)	1.64 (1.21 – 2.15)	1.85 (0.98 – 2.93)
b	Dimensionless exponent controlling curve steepness; higher values make flow rise faster with depth.	2.86 (2.41 – 3.26)	2.67 (2.29 – 3.06)	3.10 (2.32 – 3.79)
σ_{proc}	*Process variability (CFS):* other process uncertainty (e.g., turbulence, point gauge drift).	0.009 (0.001 – 0.027)	0.008 (0.001 – 0.023)	0.024 (0.002 – 0.071)
$\sigma_{Q,add}$	*Flume infiltration error (CFS):* additive discharge offset caused by minor leakage or losses through joints or print material.	0.012 (0.006 – 0.022)	0.012 (0.006 – 0.022)	0.019 (0.008 – 0.040)
c_v	*Flow-meter precision term (% of reading):* proportional error representing magnetic-flow-meter accuracy.	0.022 (0.009–0.044)	0.022 (0.009–0.044)	0.022 (0.009–0.044)

DRAFT

HPDIs for parameters a and b overlap across flumes, however, the posterior median of b for 3D Printed 2 is closest to the manufacturer value (i.e., 2.67 is closest to 2.55). The fiberglass flume exhibits a steeper posterior median curvature parameter and wider posterior intervals relative to the printed units. Similarly, while the posterior scale parameters (a) for the printed flumes cluster tightly around the expected magnitude, the fiberglass flume shows greater dispersion. However, this greater uncertainty should be interpreted in light of the smaller calibration dataset (n = 4) available for the fiberglass flume compared to each 3D printed flume (n = 6). In a Bayesian framework, posterior distribution dispersion naturally reflects both data variability and information content; with fewer observations spanning the stage range, the model has less constraint on the curvature and scale parameters. Consequently, wider HPDIs for the fiberglass unit are expected under limited sample support and do not necessarily imply inferior hydraulic performance. With additional calibration points for the fiberglass flume, one would expect its posterior intervals to contract, potentially reducing apparent differences in curvature.

A principal advantage of the Bayesian framework in this context is its explicit characterization of uncertainty in both parameters and predicted discharge. Rather than treating discharge observations as error-free, the model separates measurement error from structural hydraulic variability. Specifically, observed discharge was modeled as arising from a latent "true" discharge variable, with proportional measurement error and additive process variance estimated directly from the data. The use of latent true discharge variables further enhances interpretation of Figure 6, where filled points represent posterior estimates of the underlying hydraulic discharge after accounting for measurement noise. These points are what the parameters were calibrated to: not the raw data. This distinction allows separation of instrument variability from physical rating-curve behavior, which is not possible in a simple least-squares framework.

This approach was particularly well suited to the relatively small number of calibration points per flume (n = 4–6). Classical nonlinear regression quantifies parameter uncertainty using asymptotic

DRAFT

normal theory, which assumes that parameter estimates approach a bell-shaped distribution as sample size increases (Seber and Wild, 1989). Such asymptotic results are approximations whose accuracy depends on sufficient sample size. When data are limited, these normal approximations may not hold, and linearized standard errors derived from them can misrepresent true uncertainty or produce unstable coefficient estimates. In contrast, Bayesian inference does not rely on asymptotic normality for uncertainty quantification. Instead, it derives full posterior distributions directly from the likelihood and prior, allowing parameter uncertainty, measurement error, and process variability to be jointly propagated through the rating curve even when data are sparse (Gelman et al., 2013). This distinction is particularly valuable for laboratory flume calibration studies, where collecting large numbers of stage–discharge observations may be impractical and where overconfidence in standard errors could lead to misleading discharge estimates.

These results demonstrate that the Bayesian model not only provided stable parameter estimates under limited data but also delivered a mechanistic decomposition of uncertainty into measurement error, process variability, and structural curvature differences. These findings align with prior work demonstrating that properly fabricated 3D printed flumes can reproduce discharge relationships with high consistency and minimal between-print variability, while highlighting the additional inferential clarity gained through probabilistic modeling (Akhter et al., 2021).

Beyond parameter estimation, the practical utility of the Bayesian framework is demonstrated through the generative model aspect of the approach by creating stage–discharge predictions summarized in . Results translate the calibrated rating curves into directly usable discharge estimates across the operational depth range, with full uncertainty propagation. This is particularly important for applied water-measurement contexts, where decisions are often made at specific stage thresholds. For example, at h = 0.5 ft, both 3D printed flumes produce median predictions within 0.06 CFS of the manufacturer equation, while the fiberglass flume exhibits lower median discharge and wider uncertainty bounds. In applied irrigation and compliance settings, therefore provides a probabilistic

DRAFT

discharge lookup framework that is more informative than a single power-law curve, enabling practitioners to quantify both expected flow and associated uncertainty at any stage within the tested range.

Table 5. Posterior stage–discharge predictions generated from the hierarchical Bayesian calibration model for the two 3D printed trapezoidal flumes (3D Printed 1, 3D Printed 2) and the commercial fiberglass flume. Reported values are posterior median discharges (CFS) with 95 % high density probability intervals in parentheses, evaluated at selected depths across the operating range. The manufacturer's empirical equation is shown for reference. Uncertainty increases at higher depths where calibration data were sparser, reflecting extrapolation beyond the densest measurement region.

	Discharge (CFS)			
Stage (ft)	**3D Printed 1**	**3D Printed 2**	**Fiberglass**	**Manu. Eq.**
0.1	0.003 (0.001, 0.007)	0.003 (0.002, 0.008)	0.001 (0, 0.007)	0.004
0.2	0.018 (0.011, 0.032)	0.022 (0.014, 0.034)	0.012 (0.005, 0.03)	0.024
0.3	0.059 (0.044, 0.08)	0.065 (0.049, 0.082)	0.042 (0.024, 0.072)	0.069
0.4	0.134 (0.116, 0.155)	0.139 (0.122, 0.157)	0.104 (0.07, 0.14)	0.146
0.5	0.255 (0.237, 0.27)	0.253 (0.229, 0.274)	0.209 (0.146, 0.253)	0.259
0.6	0.43 (0.374, 0.476)	0.412 (0.353, 0.473)	0.367 (0.251, 0.463)	0.415
0.7	0.67 (0.534, 0.801)	0.621 (0.498, 0.768)	0.591 (0.366, 0.827)	0.618
0.8	0.983 (0.726, 1.252)	0.887 (0.671, 1.165)	0.896 (0.502, 1.38)	0.872

FLUME EVALUATION AFTER FIELD DEPLOYMENT

After flume uninstallation from field deployment, caliper measurements were taken of each flume dimension. Resulting values showed that the 3D printed flume's dimensions remained within a tolerance of ≤ 2 mm relative to the original large 60° V-notch trapezoidal flume design for all measurements taken (Table 6). This supports the use of PETG as a suitable plastic for flume fabrication because the material maintained its dimensional stability despite exposure to solar radiation and erosive pressure over multiple irrigation events spanning several months, validating its suitability for field applications.

DRAFT

Table 6. Dimensional comparison of Large 60° V-notch trapezoidal flume dimensions after field trails

Dimensions	Value after Field Trails (cm)	Published Value (cm) (Teledyne ISCO, 2023)
b1	5.05	5.1
b2	0.00	0.0
A	18.0	17.8
L	17.8	17.8
C	17.7	17.6
D	17.9	17.8
E	7.70	7.6
F	20.4	20.3
H	17.3	17.1
B	78.4	78.4
R	3.90	3.8
S	10.0	10.2
U	8.80	8.9
W	25.3	25.4
θ	59.9	60.0
ϕ	8.26	8.25

A minor stress fracture developed at the stilling well port junctions during uninstallation. Extracting the unit from highly compacted soil resulted in a bending moment to the attached PVC stilling well causing the minor fracture. While this fracture did not affect flume performance, it highlights an operational consideration: The relatively brittle plastic requires more careful handling during installation and extraction compared to commercial fiberglass and metal models. One significant advantage of 3D printing with PETG and other plastics, however, is the ease of repair. The fracture was easily repaired by applying 3D Gloop which chemically welds the polymer matrix. This chemical weld fully restored the structural integrity of the flumes within 24 hours, while commercial fiberglass repairs require laborious resin patching or total unit replacement. Furthermore, replacement of a 3D printed flume would cost $90 while a commercial fiberglass model would be vastly more expensive.

In addition to durability, flume performance was characterized by plotting the hydraulic response during irrigation events for visual inspection. An example of this is depicted in Figure 7, which

DRAFT

represents an irrigation event captured on September 16, 2025, that lasted approximately 15 hours. The hydrograph shows the calculated discharge by utilizing the water level data collected by the LCS and the specific rating curves derived for both 3D printed flumes, fiberglass flume, and manufacturer equation. The resulting flow data display clear characteristic flow dynamics of a furrow irrigation event, beginning with a first flush followed by a rapid spike in discharge, a high stable steady-state plateau during continuous application, and finally a steep reduction in discharge following water cutoff.

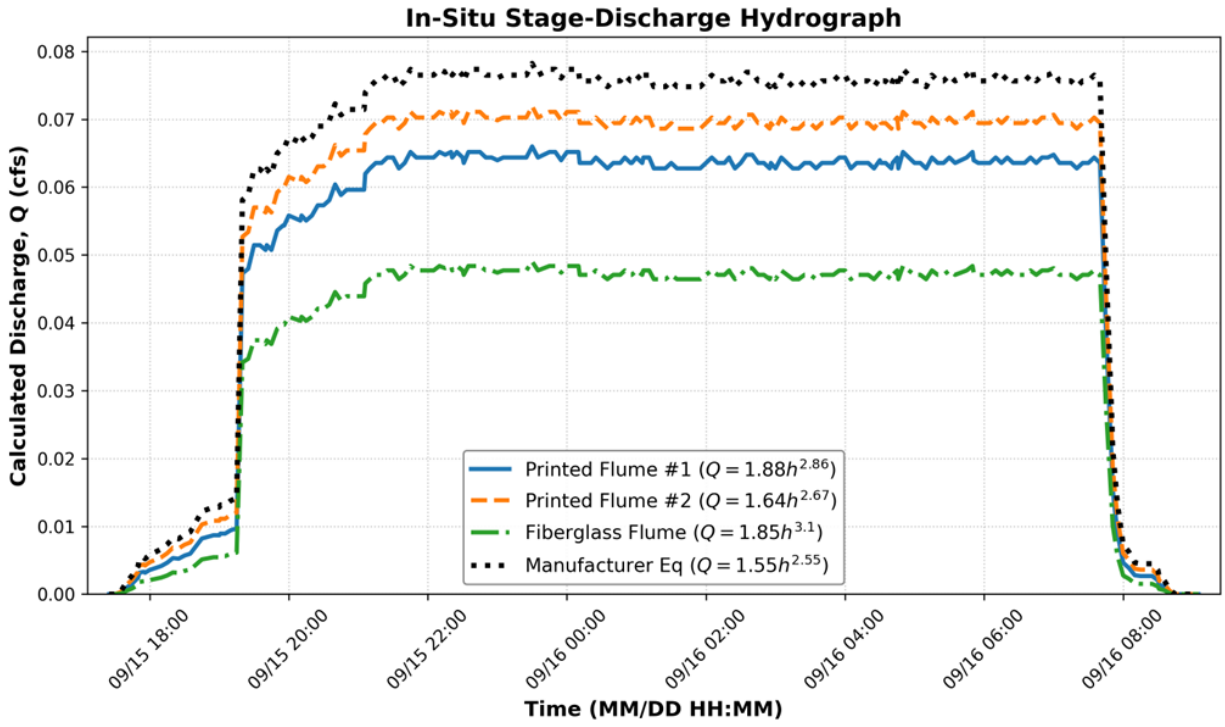


Figure 7. Discharge hydrograph for a representative 16-hour furrow irrigation event at the ARDEC test site. Continuous stage data recorded by the integrated eTape water depth sensor was converted to volumetric discharge (Q) using the empirically derived rating curves for Printed Flume #1, Printed Flume #2, Fiberglass, and manufacturer equation. Printed Flume #2 was the unit installed at ADREC during the field trial.

To illustrate the practical implications of these discharge differences, a simple mass loading example can be considered. Assuming a representative runoff concentration of 10 mg/L for a dissolved constituent such as Nitrate-Nitrogen (NO3–N), cumulative event loads can be estimated directly from the calculated volumes. Because 1 mg/L is equivalent to 1 g/m³, this corresponds to 10 g/m³, allowing

straightforward conversion from volumetric runoff to total mass export. Mass loading is defined as:

$$\text{Load} = C \times V \tag{11}$$

where C is concentration and V is cumulative runoff volume. Under this relationship, differences in estimated discharge propagate linearly into differences in total mass loading. The resulting loads for each flume configuration are summarized in Table 7.

Table 7. Example cumulative mass loading for a 16-hour irrigation event assuming a constant concentration of 10 mg/L. Percent change is calculated relative to the manufacturer's baseline.

Flume / Equation	Volume (gal)	Volume (m^3)	Load (g)	% Change from Manufacturer
Manufacturer baseline	29,954	113.4	1,134	0%
Printed Flume #2	27,418	103.8	1,038	−8.5%
Printed Flume #1	24,997	94.6	946	−16.6%
Fiberglass flume	18,431	69.8	698	−38.4%

The fiberglass flume yields a substantially lower estimated load, underpredicting cumulative export by approximately 38% relative to the manufacturer's baseline. In contrast, the 3D printed flumes produce load estimates that are considerably closer to the expected value, within approximately 8 - 17%.

From a water quality perspective, these differences are consequential. For nutrients such as NO3–N, systematic underestimation of discharge directly propagates into underestimation of mass loading, which can bias field-scale mass balances, obscure treatment effects, and affect regulatory or reporting outcomes. This reinforces that accurate stage–discharge relationships are not only important for hydrologic measurement, but are foundational to reliable water quality load estimation

FUTURE WORK

While this current study establishes the short-term feasibility of the 3D printed flume in field applications, future studies must focus on the long-term viability of these flumes. Although PETG material is known to have natural UV resistance, alternative materials, such as acrylonitrile styrene acrylate (ASA), that are known to have greater UV and impact resistance are ideal for long-term outdoor exposure. Future work will examine how discharge coefficients can be estimated or parameterized for field-customized flume geometries, enabling predictive application of critical-depth flumes without

DRAFT

requiring full empirical recalibration for each new design. Furthermore, long term environmental degradation of these 3D printed flumes raises concern regarding potential dispersion of microplastics into our water system. Future work will combine standard agricultural nutrient water quality analysis to include the quantification of potential microplastics downstream of the flume.

CONCLUSIONS

The results of this study establish large-scale, 3D printed trapezoidal flumes as an accurate and cost-effective tool for agricultural flow monitoring. Laboratory calibration determined that the 3D printed flumes are hydraulicly similar to the published stage-discharge relationship for the chosen flume design, such that the 3D flumes should be considered as a reasonable replacement when compared to a commercial fiberglass equivalent. This was investigated using a novel hierarchical Bayesian parameterization of the stage-discharge model, which quantified both parameter and measurement uncertainty simultaneously. Field deployment of the printed flume supports the structural durability of the PETG polymer for this use case; the printed flume reliably functioned as designed over a two-month deployment in a furrow irrigated system, successfully weathering environmental conditions. The flume was successfully integrated with an LCS for continuous edge-of-field water quantity and quality monitoring, supporting its use in agricultural and environmental applications. Ultimately, this research provides a defensible methodology for future flume development and *in situ* applications that reduce fabrication costs and eliminate manufacturing lead times for scalable and accessible water flow measurement.

ACKNOWLEDGMENTS

The authors would like to thank the staff at the CSU Agricultural Research, Development, and Education Center (ARDEC) and the CSU Agricultural Experiment Station for their operational support during field testing of the 3D printed flume. This project was funded through the Colorado Water Center competitive funding program to support water research, extension, and education projects.

DRAFT

Administrative support provided by the Colorado Department of Agriculture. Furthermore, the authors extend a special thanks to Alex Enriquez, Mackenize Maruca, and Abby Coney for their valuable assistance with the flume assembly, field installation, and data collection as student research assistants of the CSU Agricultural Water Quality Program.

REFERENCES

Akhter, F., McMaine, J., McLemore, A.J., Hurst, M.J., 2021a. Validation and development of discharge equations for 3D printed flumes for flow monitoring. Trans. ASABE 64, 1921–1928.

Autodesk, 2024. Autodesk Fusion.

Chow V.T., 1959. Open-channel hydraulics. McGraw-Hill, New York .

Deleon, E., Brown, A., Ladow, J., Wardle, E., Bauder, T., 2025. Programable Automated Low-Cost IoT Water Sampler. HardwareX 23, e00693. https://doi.org/https://doi.org/10.1016/j.ohx.2025.e00693

Gabry, J., Češnovar, R., Johnson, A., Bronder, S., 2025. cmdstanr: R Interface to "CmdStan."

Gelman, A., Carlin, J.B., Stern, H.S., Dunson, D.B., Vehtari, A., Rubin, D.B., 2013. Bayesian Data Analysis, Third. ed. Chapman and Hall/CRC. https://doi.org/10.1201/b16018

Gelman, A., Rubin, D.B., 1992. Inference from iterative simulation using multiple sequences. Statistical science 7, 457–472. https://doi.org/https://doi.org/10.1214/ss/1177011136

Harmel, R.D., Preisendanz, H.E., King, K.W., Busch, D., Birgand, F., Sahoo, D., 2023. A Review of Data Quality and Cost Considerations for Water Quality Monitoring at the Field Scale and in Small Watersheds. Water (Switzerland). https://doi.org/10.3390/w15173110

McElreath, R., 2020. Statistical rethinking: A Bayesian course with examples in R and Stan. Chapman and Hall/CRC.

Oertel, M., Shen, X., 2022. 3D Printing Technique for Experimental Modeling of Hydraulic Structures: Exemplary Scaled Weir Models. Water (Switzerland) 14. https://doi.org/10.3390/w14142153

Parshall, R.L.; R.C., 1921. The Venturi Flume. Fort Collins.

Posit team, 2025. RStudio: Integrated Development Environment for R.

R Core Team, 2020. R: A Language and Environment for Statistical Computing.

DRAFT

Robinson, A.R., Chamberlain, A.R., 1960. Trapezoidal flumes for open-channel flow measurement.

Seber, G.A.F., Wild, C.J., 1989. Asymptotic Theory, in: Nonlinear Regression, Wiley Series in Probability and Statistics. John Wiley & Sons, Inc., pp. 563–585. https://doi.org/https://doi.org/10.1002/0471725315.ch12

SoftFever, 2025. OrcaSlicer.

Stan Development Team, 2023. Stan Modeling Language Users Guide and Reference Manual, version 2.36.

Teledyne ISCO, 2023. Teledyne ISCO Open Channel Flow Measurement Handbook, 8th ed. Teledyne ISCO, Lincoln, NE.

Vehtari, A., Gelman, A., Simpson, D., Carpenter, B., Bürkner, P.-C., 2021. Rank-Normalization, Folding, and Localization: An Improved Rˆ for Assessing Convergence of MCMC (with Discussion). Bayesian Anal. 16. https://doi.org/10.1214/20-BA1221

NOMENCLATURE

Variables and Symbols

α = Empirical derived discharge coefficient

b = Empirical derived discharge exponent

h = Hydraulic head or flow depth (ft)

Q = Volumetric flow rate or calculated discharge (cfs)

(a) = Scale of the power-law rating curve

(b) = Curvature of the power-law rating curve

(σ_{proc}) = Process variability

($\sigma_{Q,\text{add}}$) = Additive discharge loss

(c_v) = Proportional flow-meter error

Acronyms and Abbreviations

ARDEC = Agricultural Research, Development and Educations Center

ASA = Acrylonitrile Styrene Acrylate

DRAFT

554 CAD = Computer Aided Design

555 CSU = Colorado State University

556 CSU AES = Colorado State University Agricultural Experimental Station

557 FDM = Fused Deposition Modeling

558 LCS = Low-Cost Sampler

559 HMC = Hamiltonian Monte Carlo

560 PETG = Polyethylene Terephthalate Glycol

561 USDA = United States Department of Agriculture

562 UV = Ultraviolet

563 ## APPENDIX

564 Appendix body text.

DRAFT